ESP32 TCP Socket Communication Guide

(Absolute Beginner Step-by-Step Tutorial)

Overview

This guide walks you through setting up a raw **TCP Echo Server** on an ESP32 microcontroller from scratch and testing bi-directional communication from your Windows PC using **Packet Sender**.

Prerequisites & Hardware Needed

- **ESP32 Microcontroller Board** (e.g., ESP32 Dev Module / NodeMCU)
- **Micro-USB or USB-C Data Cable** (Ensure it supports data transfer, not power-only)
- **Windows PC** connected to a 2.4 GHz Wi-Fi network

Step 1: Install Packet Sender via Command Line

1. Press `Win + X` and select **Windows Terminal** or **PowerShell**.
2. Run the following command to install **Packet Sender** via `winget`:
3. Bash

winget install packetsender

4. Once the installation completes, search for **Packet Sender** in your Start menu to confirm it is installed.

Step 2: Install and Configure Arduino IDE

1. Download and install **Arduino IDE 2.x** from arduino.cc/en/software.
2. Open Arduino IDE and go to `File > Preferences`.
3. In the **Additional Boards Manager URLs** field, paste the following link:
4. Plaintext

   https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json

5. Click **OK**.
6. Go to `Tools > Board > Boards Manager...`, search for `esp32` (by **Espressif Systems**), and click **Install**.

Step 3: ESP32 TCP Echo Server Code

1. Create a new sketch in Arduino IDE (`File > New Sketch`).
2. Replace all existing text with the following code:

[Use the file of TCP.ino]

3. Update `"YOUR_WIFI_SSID"` and `"YOUR_WIFI_PASSWORD"` with your actual Wi-Fi name and password.

Step 4: Flash the Code & Retrieve IP Address

1. Plug your ESP32 into your computer using the USB cable.
2. Select your board under `Tools > Board > esp32 > ESP32 Dev Module`.
3. Select your serial port under `Tools > Port` (e.g., `COM3` or `COM4`).
4. Click the **Upload** button (Right Arrow icon in top left).
5. Once uploaded, open the **Serial Monitor** (`Tools > Serial Monitor` or `Ctrl + Shift + M`).
6. Set the baud rate dropdown in the top-right of the Serial Monitor to **115200 baud**.
7. Press the **EN** (or **RST**) button on your ESP32 board.
8. Copy the printed IP address (e.g., `192.168.1.50`).

Step 5: Send TCP Data using Packet Sender

1. Open **Packet Sender**.
2. Fill in the top configuration fields:
    - **Name:** `ESP32 Test`
    - **ASCII:** `Hello ESP32`
    - **Address:** `<Paste ESP32 IP address your>` *(e.g., 192.168.1.50)*
    - **Port:** `8080`
    - **Protocol:** Select **TCP** from the dropdown menu.
3. **CRITICAL STEP:** Next to the ASCII input field (or in the bottom-right corner), change the **Line Ending** dropdown from `None` to **\n** or **\r\n (CR+LF)**. *(The ESP32 code relies on `\n` to mark the end of a line).*
4. Click **Send** on the right side.

Step 6: Verify the Output

- **Packet Sender Window:** In the bottom log panel, you will see an outgoing blue **TCP** packet sent, immediately followed by a green incoming **TCP** packet displaying:
- Plaintext

ESP32 Echo: Hello ESP32

Arduino Serial Monitor: Will display:

New client connected!
Received: Hello ESP32

Troubleshooting & Key Tips

- **2.4 GHz Network Only:** ESP32 chips do not support 5 GHz Wi-Fi networks. Ensure both your PC and ESP32 are on a 2.4 GHz SSID.
- **No Response in Packet Sender:** Verify that you selected `\n` or `\r\n` line ending. Without a line break, `client.readStringUntil('\n')` will keep waiting.
- **Connection Timeout / Refused:** Ensure your PC firewall isn't blocking outgoing traffic on port `8080` and that router AP isolation is turned off.